QUALITY INFORMATION DOCUMENT

For
OMI_CLIMATE_SI_BALTIC_volume

Issue: 4.0

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RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	2019/02/20	all	Creation of the document	U. Raudsepp I. Maljutenko L. Axell	P. Lagemaa
2.0	2019/10/10	all	Update to 2018 data	U. Raudsepp I. Maljutenko	P. Lagemaa
2.1	2021/10/01	all	Remove temporal extension specific details	U. Raudsepp I. Maljutenko	P. Lagemaa
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1) INTRODUCTION

a) Product Name

OMI_CLIMATE_SI_BALTIC_volume.

b) Indicator Evaluation Method

This document provides to CMEMS users information on the quality of the daily Sea Ice Volume (SIV) index for the Baltic Sea.

The data are daily SIV values are derived from daily sea ice charts. The ice charts are produced by manually drawing polygons of ice, where each polygon has a set of attributes, including sea ice concentration and level ice thickness. The original data used to produce the charts are satellite images (SAR) as well as in situ observations (reports) from ice breakers in the Baltic Sea. SIV is computed as the sum of the area of the grid cells that have a SIC value larger than 15% multiplied with corresponding sea ice thickness.

c) Copernicus Marine products used as input for the computation

Product name (and Type)	Documentation
SEAICE_BAL_PHY_L4_MY_011_019 (Satellite observations)	QUID: https://documentation.marine.copernicus.eu/QUID/CMEMS-SEAICE-QUID-011-004-011-019.pdf PUM: https://documentation.marine.copernicus.eu/PUM/CMEMS-SI-PUM-011-019.pdf

2) SCIENTIFIC VALIDATION

a) Baltic Sea Ice Volume

The accuracy of the index is directly dependent on the accuracy of the sea ice charts (product reference SEAICE_BAL_PHY_L4_MY_011_019) from which it is calculated. The 1 km grids of the ice concentration and mean ice thickness from the FMI and SMHI ice charts were compared. For ice concentration the SMHI and FMI ice chart datasets are closer to each other at low and high concentrations, and the standard deviation is higher for the intermediate values. There are no ground-truths available for validating a sea ice volume index and therefore it is not possible to assign a direct accuracy number to this product. However, the ice concentration comparison with the radiometer product of University of Bremen has shown good agreement with these products. Based on that comparison the accuracy of the concentration of FMI ice charts has been estimated to be ±20%. Validation results with respect to the ice breaker ice thickness observations is estimated to be ±10cm (Karvonen, J., et al., 2024). From those uncertainties the uncertainty of ice volume is calculated for each day (Figure 1). In average, the uncertainty of ice volume is ±5.8km^3.

Qualitative comparison of variations of the maximum ice thickness in the Baltic Sea is made with the plot published by European Environmental Agency with the data provided by Finnish Meteorological Institute). Scientific evaluation can be found in the CMEMS Ocean State Report (Samuelsen et al., 2016; Samuelsen et al., 2018).

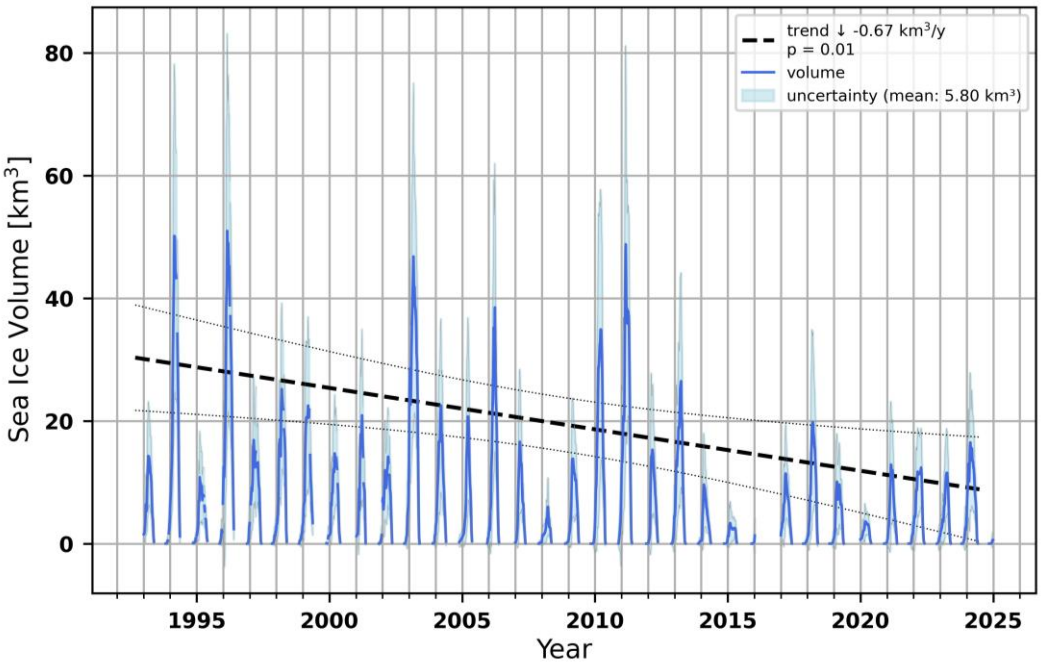

Figure 1. Time series of the area integrated daily sea-ice volume and trend, both with corresponding uncertainties, for the Baltic Sea in 1993–2025. Data is smoothed using 7-day window moving average filter.

3) REFERENCES

Samuelsen et al., 2016: Sea ice. In: The Copernicus Marine Environment Monitoring Service Ocean State Report, issue 1, Journal of Operational Oceanography, Volume 9, 2016 - Issue sup2: The Copernicus Marine Environment Monitoring Service Ocean, http://dx.doi.org/10.1080/1755876X.2016.1273446.

Samuelsen et al., 2018: Sea Ice. In: The Copernicus Marine Environment Monitoring Service Ocean State Report, issue 2, Journal of Operational Oceanography, in press.

Karvonen, J., et al., 2024: QUALITY INFORMATION DOCUMENT for Sea Ice Thematic Assembly Centre products SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_004 SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_011 SEAICE_BAL_PHY_L4_MY_011_019. https://documentation.marine.copernicus.eu/QUID/CMEMS-SEAICE-QUID-011-004-011-019.pdf